



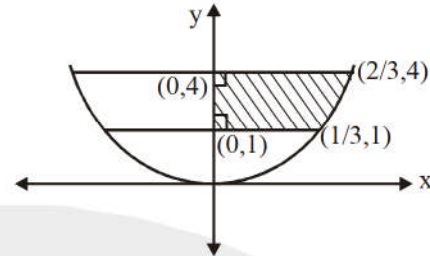
WORK SHEET - 23

AREA UNDER THE CURVE + DIFFERENTIAL EQUATION

Q.1 (B)

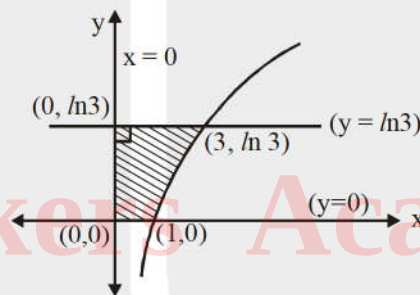
$$y = 9x^2, x = 0, y = 1, y = 4$$

$$\begin{aligned} \text{Area of shaded region} &= \int_1^4 \left(\frac{\sqrt{y}}{3} \right) dy \\ &= \frac{2}{9} \left(y^{\frac{3}{2}} \right) \Big|_1^4 \\ &= \frac{2}{9} \times (2^3 - 1^3) = \frac{14}{9} \end{aligned}$$



Q.2 (C)

$$y = \ln x, y = 0, y = \ln 3, x = 0$$

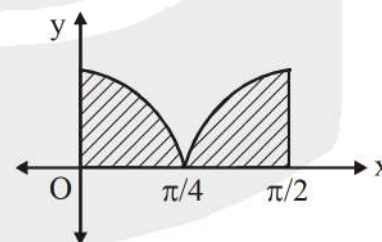


$$\text{Area} = \int_{y=0}^{y=\ln 3} e^y dy = e^y \Big|_0^{\ln 3} = e^{\ln 3} - e^0 = 3 - 1 = 2 \quad \text{Ans.}$$

Q.3 (B)

$$y = |\cos x - \sin x|, 0 \leq x \leq \frac{\pi}{2}$$

$$\begin{aligned} \text{Area} &= 2 \int_0^{\pi/4} (\cos x - \sin x) dx \quad (\text{By symmetry}) \\ &= 2 [\sin x + \cos x]_0^{\pi/4} \\ &= 2 [\sqrt{2} - 1] = 2\sqrt{2} - 2 \quad \text{Ans.} \end{aligned}$$



Q.4 (A)

$$(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$$



$$\therefore \frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{4x^2}{1+x^2}$$

$$\text{I.F.} = e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = 1+x^2$$

$$\therefore y(1+x^2) = \int (1+x^2) \left(\frac{4x^2}{1+x^2} \right) dx = \frac{4x^3}{3} + C$$

$$\therefore 3y(1+x^2) = 4x^3 + C$$

It passes through (0,0)

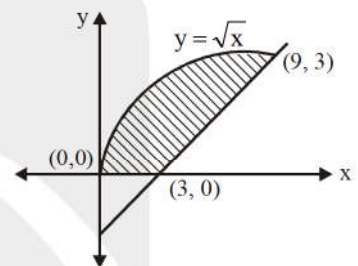
$$\therefore C = 0$$

$$\therefore 3y(1+x^2) = 4x^3 \text{ Ans.}$$

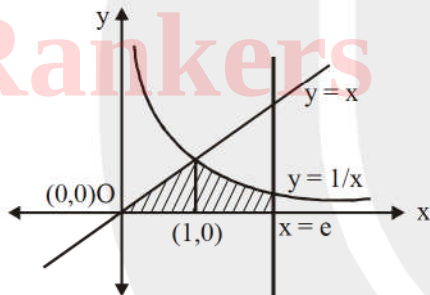
Q.5 (D)

$y = \sqrt{x}$, $2y - x + 3 = 0$, $y = 0$, Ist quadrant

$$\begin{aligned} \text{Area} &= \int_0^9 \sqrt{x} dx - \int_3^9 \left(\frac{x-3}{2} \right) dx = \frac{2}{3} x^{\frac{3}{2}} \Big|_0^9 - \left(\frac{x^2}{4} - \frac{3x}{2} \right) \Big|_3^9 \\ &= \frac{2}{3} [27] - \left(\frac{81}{4} - \frac{27}{2} - \frac{9}{4} + \frac{9}{2} \right) = 18 - 9 = 9 \text{ Ans.} \end{aligned}$$



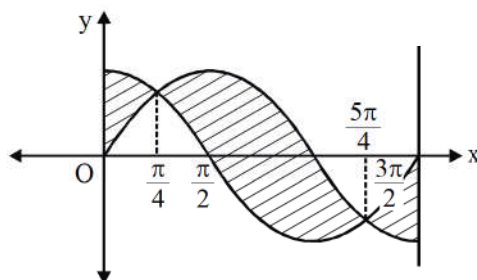
Q.6 (B)



$$\text{Area} = \int_1^e \frac{1}{x} dx + \frac{1}{2} \times (1 \times 1) = \frac{1}{2} + \ln x \Big|_1^e = \frac{1}{2} + 1 = \frac{3}{2} \text{ Ans.}$$

Q.7 (D)

$$y = \cos x, y = \sin x, x = 0, x = \frac{3\pi}{2}$$



$$\begin{aligned}\text{Area} &= 2 \int_0^{\pi/4} (\cos x - \sin x) + \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) = 2(\sin x + \cos x) \Big|_0^{\pi/4} + (-\cos x - \sin x) \Big|_{\pi/4}^{5\pi/4} \\ &= 2\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1\right) + (\cos x + \sin x) \Big|_{\pi/4}^{5\pi/4} = 2(\sqrt{2} - 1) + (2\sqrt{2}) = 4\sqrt{2} - 2 \quad \text{Ans.}\end{aligned}$$

Q.8 (D)

$$\frac{dy}{dx} = \frac{x+y}{x} = 1 + \frac{y}{x}$$

$$\therefore \frac{dy}{dx} - \frac{y}{x} = 1$$

$$\text{I.F.} = \frac{1}{x}$$

$$\therefore \frac{y}{x} = \int \frac{1}{x} \cdot 1 \, dx - \ln x + C$$

$$\therefore y = x \ln x + cx$$

$$y(1) = 1$$

$$\therefore 1 = C$$

$$\therefore C = 1$$

$$\therefore y = x \ln x + x \quad \text{Ans.}$$

Q.9 (D)

$$x + 2y^2 = 0$$

$$x + 3y^2 = 1; \Rightarrow y^2 = \frac{-1}{3}(x-1)$$

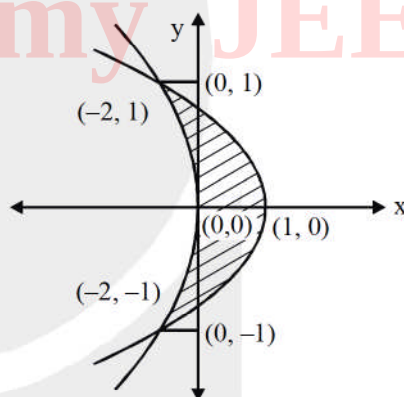
$$\Rightarrow -2y^2 = 1 - 3y^2 \Rightarrow y^2 = 1 \Rightarrow y = \pm 1$$

$$\therefore x = -2$$

$$\Rightarrow A(-2, 1), B(-2, -1)$$

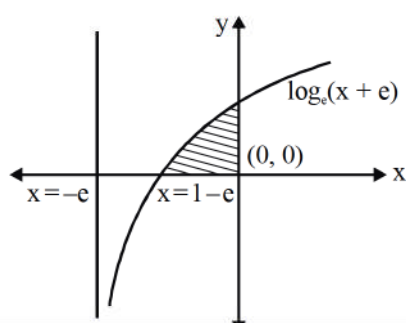
$$\therefore \text{Required area} = 2 \int_0^1 ((1-3y^2) - (-2y^2)) dy$$

$$= 2 \int_0^1 (1-y^2) dy = 2 \left(y - \frac{y^3}{3} \right) \Big|_0^1 = 2 \left(1 - \frac{1}{3} \right) = \frac{4}{3} \quad \text{Ans.}$$



Q.10 (A)

$$y = \log_e(x+e)$$



$$\text{Area} = \int_{1-e}^0 \log_e(x+e) dx$$

$$= [(x+e) \log_e(x+e) - (x+e)]_{1-e}^0 = (e-e) - [-(1-e+e)] = 1 \text{ Ans.}$$

Q.11 (D)

$$y^2 = 4x; \quad x^2 = 4y$$

$$x = 4; \quad y = 4$$

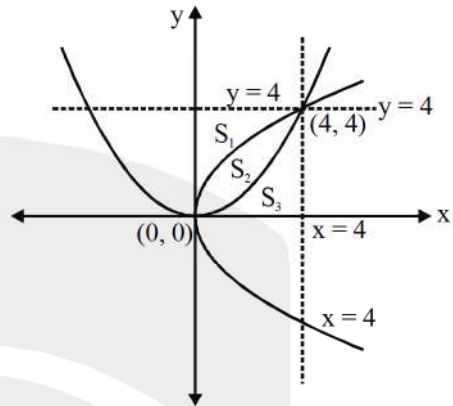
$$x = 0; \quad y = 0$$

$$S_3 = \int_0^4 \frac{x^2}{4} dx = \frac{x^3}{3 \cdot 4} = \frac{64}{3 \cdot 4} = \frac{16}{3}$$

$$S_1 = \int_0^4 \frac{y^2}{4} dy = \frac{y^3}{12} = \frac{64}{12} = \frac{16}{3}$$

$$S_2 = \int_0^4 \left[2\sqrt{x} - \frac{x^2}{4} \right] dx = 2 \cdot \frac{2}{3} x^{\frac{3}{2}} \Big|_0^4 - \frac{x^3}{12} \Big|_0^4$$

$$= \frac{32}{3} - \frac{16}{3} = \frac{16}{3} \Rightarrow S_1 = S_2 = S_3 \text{ Ans.}$$



Q.12 (A)

$$y = c_1 x + \sqrt{c_1}$$

$$\frac{dy}{dx} = c_1$$

$$\therefore y = \frac{dy}{dx} x + \sqrt{\frac{dy}{dx}}$$

$$\therefore \left(y - \frac{xdy}{dx} \right) = \sqrt{\frac{dy}{dx}}$$

$$\text{or} \quad \left(y - \frac{xdy}{dx} \right)^2 = \frac{dy}{dx}$$

$\therefore$ order is 1 and degree is 2 Ans.

Q.13 (C)

$$y = c_1 e^{2x} + c_2 e^x + c_3 e^{-x}$$

Divide by e^{2x}

$$\therefore y e^{-2x} = c_1 + c_2 e^{-x} + c_3 e^{-3x}$$

Differentiate w.r.t. 'x'

$$\therefore y (-2e^{-2x}) + e^{-2x} y' = -c_2 e^{-x} - 3c_3 e^{-3x}$$

Multiply by e^x ,

$$y(-2e^{-x}) + e^{-x} y' = -c_2 - 3c_3 e^{-2x}$$

Differentiate with respect to x,

$$y(2e^{-x}) + (-2e^{-x})y' + e^{-x}y'' + y'(-e^{-x}) = 6c_3 e^{-2x}$$

Multiply by e^{2x} , we get

$$\begin{aligned}\therefore y(2e^x) - (3e^x y') + (e^x y'') - y'e^x &= 6c_3 \\ \therefore 2ye^x + 2e^x y' - 3e^x y'' - 3y'e^x + e^x y''' + y''e^x &= 0 \\ \therefore y'''e^x - 2y''e^x - e^x y' + 2ye^x &= 0 \\ \therefore y''' - 2y'' - y' + 2y &= 0 \\ a = -2, b = -1, c = 2\end{aligned}$$

$$\therefore \frac{a^3 + b^3 + c^3}{abc} = \frac{-8 + 8 - 1}{4} = \frac{-1}{4} \text{ Ans.}$$

Q.14 (C)

$$\begin{aligned}f: \mathbb{R} \rightarrow \mathbb{R}, g: \mathbb{R} \rightarrow \mathbb{R}, \quad f''(x) &= g''(x) \\ \therefore f'(x) &= g'(x) + c \\ f'(1) &= g'(1) + c \\ \therefore 2 &= 4 + c; \quad \therefore c = -2 \\ \therefore f'(x) &= g'(x) - 2 \\ \therefore f(x) &= g(x) - 2x + c \\ f(2) &= g(2) - 4 + c \\ \therefore 3 &= 9 - 4 + c; \quad \therefore c = -2 \\ \therefore f(x) &= g(x) - 2x - 2 \\ f(4) - g(4) &= -2(4) - 2 = -10 \text{ Ans.}\end{aligned}$$

Q.15 (C)

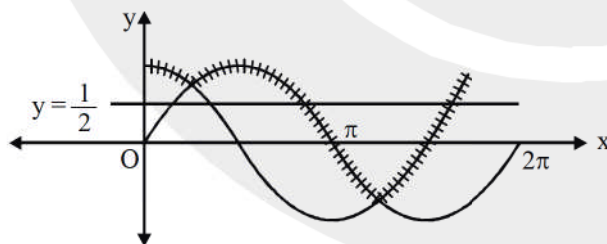
$$y = x - ax^2, ay = x^2$$

$$\text{Bounded area} = \int_0^{\frac{a}{1+a^2}} \left(x - ax^2 - \frac{x^2}{a} \right) dx = \frac{a^2}{6(1+a^2)^2} = \frac{1}{6\left(a + \frac{1}{a}\right)^2} \leq \frac{1}{24}$$

$\therefore$ Value of a for which area is maximum is 1. Ans.

Q.16 (C)

$$\text{Max.}(\sin \theta, \cos \theta) = \frac{1}{2}, \theta \in (-2\pi, 5\pi)$$



Q.17 (C)

$$y = n \ln x$$

$$A(n) = \int_1^e y dx = \int_1^e n \ln x dx = n \left[\int_1^e \ln x dx \right] = n \left[(\ln x - x)_1^e - \int_1^e x \cdot \frac{1}{x} dx \right] = n[e - [e - 1]] = n$$

$$A(n) = n,$$

$$\text{Now, } A(n) + nA(n-1) = n + n(n-1) = n + n^2 - n = n^2$$



Q.18 (D)

$$f(x) = ax^2 + bx + c$$

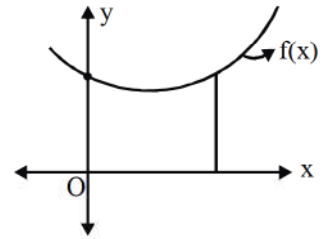
$$a \in \mathbb{R}^+, b^2 - 4ac < 0$$

$$\text{Now, } \int_0^1 f(x) dx = \int_0^1 (ax^2 + bx + c) dx$$

$$= \left[\frac{ax^3}{3} + \frac{bx^2}{2} + cx \right]_0^1 = \left[\frac{a}{3} + \frac{b}{2} + c \right] = \frac{1}{6} [2a + 3b + 6c]$$

By simplifying option

$$= \frac{1}{12} (5f(1) - f(-1) + 8f(0))$$



Q.19 (A)

$$[x + y + 1] = [x] \Rightarrow [x + y] + 1 = [x]$$

$$\text{For } x \in (0, 1) \Rightarrow [x] = 0$$

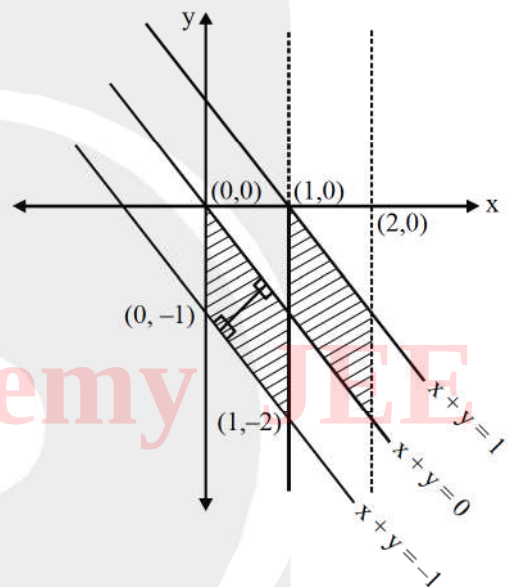
$$\Rightarrow [x + y] = -1 \Rightarrow -1 \leq x + y < 0$$

$$\text{For } x \in (1, 2) \Rightarrow [x] = 1$$

$$\Rightarrow [x + y] = 0 \Rightarrow 0 \leq x + y < 1$$

$$\therefore \text{Required shaded area} = 2 (\text{length of base}) \times (\text{height})$$

$$= 2(\sqrt{2}) \left(\frac{1}{\sqrt{2}} \right) = 2 \text{ square units}$$



Q.20 (A)

$$y = \frac{2t}{1+t^2}, \quad x = \frac{1-t^2}{1+t^2}$$

$$\text{Put } t = \tan \theta$$

$$y = \frac{2 \tan \theta}{1 + \tan^2 \theta}, \quad x = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$y = \sin 2\theta; \quad x = \cos 2\theta$$

$$\therefore \text{Squaring and adding}$$

$$x^2 + y^2 = 1$$

It is a circle.

$$\text{Area} = \pi r^2 \Rightarrow \pi(1)^2 \Rightarrow \pi$$

Q.21 (D)

$$\frac{dy}{dx} = \frac{y(\sin x - y)}{\cos x} \Rightarrow \frac{dy}{dx} + (-\tan x)y = -(\sec x)y^2$$

$$\Rightarrow y^{-2} \frac{dy}{dx} + \frac{(-\tan x)}{y} = -(\sec x) \quad (\text{Bernoulli's differential equation})$$

$$\text{Put } \frac{1}{y} = t$$

$$\Rightarrow \frac{-1}{y^2} \frac{dy}{dx} = \frac{dt}{dx} \Rightarrow \frac{dt}{dx} + (\tan x) t = \sec x \quad (\text{Linear differential equation})$$

$$\therefore \text{I.F.} = e^{\int (\sec x)} = \sec x$$

So, general solutions

$$t(\sec x) = \int \sec^2 x \, dx + c$$

$$\frac{\sec x}{y} = (\tan x + c) \Rightarrow \sec x = (\tan x + c)y$$

Q.22 (C)

$$\frac{dy}{dx} = \frac{y}{x} \cdot \left(\log \left(\frac{y}{x} \right) + 1 \right)$$

$$\text{Put } \frac{y}{x} = v$$

$$\Rightarrow v + x \frac{dv}{dx} = v \log v + v \Rightarrow \int \frac{dv}{v \log v} = \int \frac{dx}{x}$$

$$\log(\log v) = \log x + \log c \Rightarrow \log \left(\frac{y}{x} \right) = cx \quad \text{Ans.}$$

Q.23 (D)

$$y = c_1 e^{c_2 x} \quad \dots\dots(1)$$

$$\frac{dy}{dx} = (c_1 e^{c_2 x}) \cdot c_2 \quad \dots\dots(2)$$

$$\frac{d^2 y}{dx^2} = (c_1 e^{c_2 x}) \cdot c_2^2 \quad \dots\dots(3)$$

$$\therefore \frac{d^2 y}{dx^2} = c_2 \frac{dy}{dx} \quad [\text{using (2) in (3)}]$$

$$\Rightarrow y'' = \left(\frac{y'}{y} \right) y' \quad [\text{using (1) and (2) } c_2 = \frac{y'}{y}]$$

$$\Rightarrow yy'' = (y')^2 \quad \text{Ans.}$$